



Habib University
shaping futures

Habib University

OBJECT-ORIENTED PROGRAMMING FALL 2025

Week 02 - Part 1

Arrays and C-Strings





THIS WEEK'S AGENDA

- 01. C++ as a compiled high-level language
- 02. OOP Paradigm vs Procedural Paradigms
- 03. Revisiting loops (for-loop, while-loop and do-while-loop)
- 04. Increment Operators (pre and post)
- 05. Switch-case Statements
- 06. Arrays in C++ (C-String and Array of Strings)
- 07. Revisiting functions in C++
- 08. Passing Arrays to functions





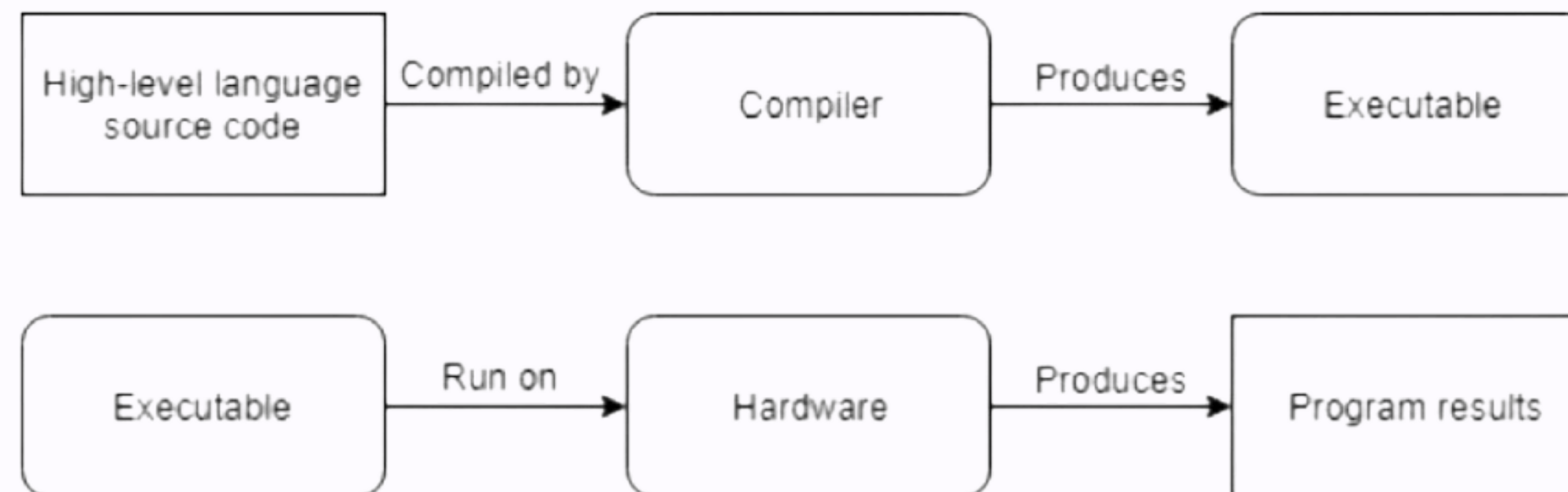
C++ AS A COMPILED HIGH-LEVEL LANGUAGE





C++ IS COMPILED

- C++ is a compiled language which means that the source code is compiled into machine readable programs once with the machine language instructions saved in a separate executable file.
- When the executable file is run, the translation step is skipped and the program runs much faster than if it were written using a interpreted language.





OOP PARADIGM VS PROCEDURAL PARADIGMS





OOP VS PROCEDURAL

Procedural Paradigm	1. Programs are written as functions. 2. Functions share global data. 3. Functions transform data from one form to another. 4. Top-down approach in program design.
Object Oriented Paradigm	1. Programs are divided into objects, 2. Data is hidden and can't be accessed by external functions. 3. Objects communicate with each other through functions. 4. Bottom-up approach in program design.





REVISITING LOOPS (FOR, WHILE, DO-WHILE)





FOR-LOOP

Lets demo this live!
We'll also look at the increment operator '++'





WHILE-LOOP

```
1  #include <iostream>
2
3  int main() {
4
5      int i = 0;
6      while (i < 5) {
7          std::cout << "i = " << i << std::endl;
8          i++;
9      }
10
11     return 0;
12 }
```

Lets demo this live!





DO-WHILE-LOOP

```
do {  
    // code block to be executed  
}  
while (condition);
```

Task: Try coding a simple do-while loop which prints “hello world” 5 times.





SWITCH-CASE STATEMENTS





SWITCH CASE STATEMENTS

- A special construct in C++ that uses multiple if statements
- The syntax is:
 - The switch expression is evaluated once.
 - The value of the expression is compared with the value of each case.
 - If there is a match, the associated block is executed.
 - The break keyword is used to skip subsequent cases.
 - The default keyword is used to jump to the default case.

```
switch(expression) {  
    case x:  
        // code block  
        break;  
    case y:  
        // code block  
        break;  
    default:  
        // code block  
}
```





SWITCH CASE STATEMENTS

- The expression must evaluate to an integer or constant values.
- Floating point or double precision values are **not allowed**.
- Enumerated values (**enum**) may also be used for the expression.
 - *What is enum?*

Lets demo this live!

